# A Certified Lower Bound for Lebesgue's Universal Cover Problem

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**Abstract**

Lebesgue's universal cover problem asks for a planar set of least possible area that contains a congruent copy of every planar set of diameter one. We work in the convex Brass–Sharifi three-test-set framework, where the test sets are a disk, an equilateral triangle, and a regular pentagon of diameter one. For the normalized placement problem, we give a finite certificate proving that the hull-area function satisfies

$$A(v) \geq 0.83201$$

on the admissible normalized domain.

The proof is a finite-cover argument. The admissible domain is covered by finitely many parameter domains, and each domain carries a local lower-bound certificate. On the witness domains, the local bound is obtained from an inner-witness polygon construction: witness points lying in the three test sets form an inner polygon whose area is bounded below by interval orientation and shoelace estimates. Since this polygon lies inside the corresponding convex hull, its area gives a lower bound for the hull area. Combining the local inequalities with the finite cover yields

$$\alpha_{\mathrm{cvx}} \geq 0.83201.$$

## 1 Introduction

A planar set is a universal cover if it contains a congruent copy of every planar set of diameter one. In this paper the covering set is required to be convex. Let

$$\mathcal{U}_{\mathrm{cvx}} = \left\{ K \subset \mathbb{R}^2 : \begin{array}{l} K \text{ is convex, and for every } S \subset \mathbb{R}^2 \text{ with } \operatorname{diam}(S) = 1, \\ \text{there is a rigid motion } g \text{ such that } g(S) \subseteq K \end{array} \right\}.$$

Set

$$\alpha_{\mathrm{cvx}} = \inf_{K \in \mathcal{U}_{\mathrm{cvx}}} \operatorname{area}(K).$$

All lower-bound statements below concern this convex quantity. Background on the wider covering problem is given by Baez, Bagdasaryan, and Gibbs [2]; a small convex covering construction is described by Gibbs [3].

Brass and Sharifi introduced the three-test-set framework used here [1]. The test sets are a disk, an equilateral triangle, and a regular pentagon of diameter one. Their normalization turns the relative placement problem into a finite-dimensional problem for the area of the convex hull of these three sets. The present work certifies the threshold

$$\tau = 0.83201$$

within this normalized convex setting.

The structure of the proof is as follows. First, a convex universal cover contains congruent copies of the three test sets, and convexity forces it to contain their convex hull. Second, the Brass–Sharifi normalization represents the hull by a parameter $v$ in an admissible domain $\Omega_{\mathrm{adm}}$. Third, the certificate covers $\Omega_{\mathrm{adm}}$ by finitely many domains and verifies a local lower bound on each one. Fourth, the finite-cover implication gives $A(v) \geq \tau$ throughout $\Omega_{\mathrm{adm}}$, and taking the infimum over convex universal covers gives $\alpha_{\mathrm{cvx}} \geq \tau$.

*Outline.* Section 2 defines the normalized three-test-set problem. Section 3 states the finite certificate model and the finite-cover implication used by the proof. Section 4 explains the sources of local lower bounds. Section 5 gives the inner-witness polygon construction. Section 6 describes the interval orientation and shoelace estimates. Section 7 proves the certified lower-bound theorem and the convex consequence. Section 8 concludes and records the scope of the statement. Appendices A and B summarize the finite certificate data used by the proof.

# 2 The normalized three-test-set problem

## 2.1 The three test sets

Let $C$, $T$, and $P_5$ be, respectively, the disk, the equilateral triangle, and the regular pentagon of diameter one. These are necessary test sets for a lower-bound argument: every universal cover contains a congruent copy of each of them. Thus, if $K \in \mathcal{U}_{\mathrm{cvx}}$, then $K$ contains three such copies; since $K$ is convex, it also contains their convex hull.

## 2.2 Normalized placements

Following the Brass–Sharifi normalization, the disk is fixed at the origin and the orientation of the triangle is fixed. The regular pentagon is taken in a fixed reference orientation; its rotational symmetry permits the pentagon angle to be recorded in a fundamental interval. Let $R_\rho$ denote rotation by angle $\rho$. Let

$$v = (\rho, x_3, y_3, x_5, y_5), \qquad u_3 = (x_3, y_3), \qquad u_5 = (x_5, y_5).$$

Here $u_3$ is the triangle translation, $\rho$ is the pentagon rotation, and $u_5$ is the pentagon translation.

Set

$$X(v) = C \cup (T + u_3) \cup (R_\rho P_5 + u_5), \qquad H(v) = \operatorname{conv} X(v),$$

and set

$$A(v) = \operatorname{area}(H(v)).$$

Figure 1 illustrates the normalized placement.

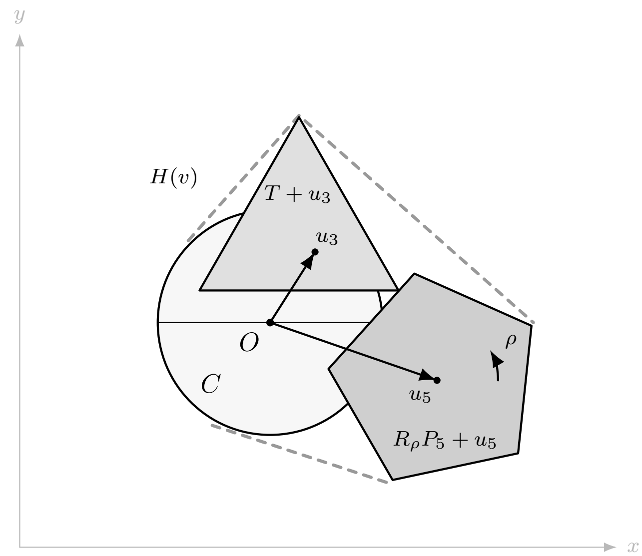


Figure 1: Schematic normalized placement of the three Brass–Sharifi test sets. The disk $C$ is fixed at the origin, the triangle is translated by $u_3 = (x_3, y_3)$, and the pentagon is rotated by $\rho$ and translated by $u_5 = (x_5, y_5)$. The enclosing curve represents $H(v) = \operatorname{conv}(C \cup (T + u_3) \cup (R_\rho P_5 + u_5))$.

### 2.3 The admissible domain and normalization principle

Denote by $\Omega_{\mathrm{adm}}$ the admissible normalized placement domain in the Brass–Sharifi framework. We use the following normalization principle from that framework: for every $K \in \mathcal{U}_{\mathrm{cvx}}$, there exists $v \in \Omega_{\mathrm{adm}}$ such that

$$H(v) \subseteq K.$$

Consequently, a uniform lower bound for $A(v)$ on $\Omega_{\mathrm{adm}}$ gives a lower bound for $\operatorname{area}(K)$ for every convex universal cover $K$.

The certificate proves

$$A(v) \geq \tau \qquad (v \in \Omega_{\mathrm{adm}}).$$

The convex universal-cover consequence is proved in Section 7.

## 3 The finite certificate model

### 3.1 Certificate cover

Let $\mathcal{F}$ be the finite cover supplied by the certificate. Its mathematical role is the inclusion

$$\Omega_{\mathrm{adm}} \subseteq \bigcup_{B \in \mathcal{F}} B,$$

where each $B \in \mathcal{F}$ is a parameter domain. The symbol $\mathcal{F}$ always denotes the cover family, while $B$ denotes one of its domains.

### 3.2 Local lower-bound assertions

For each $B \in \mathcal{F}$, the certificate supplies a value $L_B$ such that

$$A(v) \geq L_B \qquad (v \in B),$$

and it checks

$$L_B \geq \tau.$$

These two inequalities are the local facts used by the final aggregation.

### 3.3 Evidence records and interval enclosures

The certificate contains finite evidence for the local assertions. Supporting local records certify lower bounds on part of the cover. Witness records certify the witness domains, as described in Sections 5 and 6. Individual evidence records mean that each local assertion used in the cover is tied to a checked record, rather than only to an aggregate count.

The numerical records use outward-rounded interval arithmetic. Each coordinate, determinant, and shoelace expression is evaluated in an interval enclosure rounded outward, so a recorded lower endpoint is a rigorous lower bound under the stated interval model. This is the standard principle of interval arithmetic; see Moore [4].

The sizes of the finite cover and the supporting evidence records are summarized in Table A.1. The finite certificate conditions used in the final aggregation are listed in Table A.2. These tables summarize checked certificate data; the proof uses the corresponding local inequalities, not the table entries as independent assumptions.

### 3.4 Finite-cover implication

The finite-cover implication is elementary but central. If $\Omega_{\mathrm{adm}}$ is covered by $\mathcal{F}$, and if each domain $B \in \mathcal{F}$ carries a local inequality $A(v) \geq \tau$, then the same inequality holds throughout $\Omega_{\mathrm{adm}}$. Section 7 states this as a proposition and applies it to the certificate.

## 4 Sources of local lower bounds

Let $\mathcal{F}_{\mathrm{wit}} \subseteq \mathcal{F}$ be the subfamily of witness domains, and set

$$\mathcal{F}_{\mathrm{sup}} = \mathcal{F} \setminus \mathcal{F}_{\mathrm{wit}}.$$

For $B \in \mathcal{F}_{\mathrm{sup}}$, the value $L_B$ is supplied by supporting interval records and the associated individual evidence records. For $B \in \mathcal{F}_{\mathrm{wit}}$, the value $L_B$ is realized by a witness lower bound $L_B^{\mathrm{wit}}$.

### 4.1 Supporting domains

On the supporting domains $B \in \mathcal{F}_{\mathrm{sup}}$, the certificate supplies local lower bounds through interval records that are tied to individual evidence rows. Once the value $L_B$ is certified, these domains enter the proof only through the local inequalities of Section 3.2.

### 4.2 Witness domains

On the witness domains $B \in \mathcal{F}_{\mathrm{wit}}$, the local value is obtained from an inner-witness polygon. In this case the certificate verifies

$$L_B = L_B^{\mathrm{wit}}, \qquad A(v) \geq L_B^{\mathrm{wit}} \qquad (v \in B),$$

and the interval estimates of Section 6 verify $L_B^{\mathrm{wit}} \geq \tau$. The witness-domain counts and interval lower-bound minima are summarized in Tables B.1 and B.2.

# 5 The inner-witness polygon construction

## 5.1 Witness point families

Fix $B \in \mathcal{F}_{\mathrm{wit}}$. A witness family on $B$ is a finite set

$$Q_B(v) = \{q_1(v), \dots, q_{m_B}(v)\}$$

with

$$q_i(v) \in X(v) \qquad (1 \le i \le m_B,\ v \in B).$$

Each witness point has recorded provenance: it is a translated vertex of $T$, a rotated-translated vertex of $P_5$, or a certified point of $C$. The triangle witness coordinates are affine in the translation variables. The pentagon witness coordinates are trigonometric-translation functions of $\rho$ and $u_5$. The disk witnesses carry interval containment certificates. In all cases, the coordinate enclosures on $B$ are evaluated by outward-rounded interval arithmetic.

## 5.2 Containment in the hull

Set

$$W_B(v) = \operatorname{conv} Q_B(v).$$

Since $Q_B(v) \subseteq X(v) \subseteq H(v)$, convexity gives

$$W_B(v) \subseteq H(v).$$

Thus the area of $W_B(v)$ is a lower bound for the hull area.

## 5.3 The inner-witness lower-bound lemma

**Lemma 1** (Inner-witness lower bound)**.** *Suppose that $Q_B(v) \subseteq X(v)$ for all $v \in B$, and that*

$$\operatorname{area}(W_B(v)) \ge L_B^{\mathrm{wit}} \qquad (v \in B).$$

*Then*

$$A(v) \ge L_B^{\mathrm{wit}} \qquad (v \in B).$$

*Proof.* For $v \in B$, the containment $Q_B(v) \subseteq X(v) \subseteq H(v)$ implies

$$W_B(v) = \operatorname{conv} Q_B(v) \subseteq \operatorname{conv} X(v) = H(v).$$

Area is monotone under inclusion for planar measurable sets, so

$$A(v) = \operatorname{area}(H(v)) \ge \operatorname{area}(W_B(v)) \ge L_B^{\mathrm{wit}}.$$

□

# 6 Interval orientation and shoelace estimates

## 6.1 Coordinate enclosures

Write

$$q_i(v) = (x_i(v), y_i(v)), \qquad 1 \le i \le m_B.$$

The certificate evaluates interval enclosures for these coordinate functions on each witness domain $B$. The enclosures are rounded outward at every operation. Consequently, all determinant and shoelace lower endpoints used below are conservative.

### 6.2 Cyclic order and determinant lower bounds

For consecutive triples in the prescribed cyclic order, define

$$\Delta_{B,i}(v) = \det\bigl(q_{i+1}(v) - q_i(v),\, q_{i+2}(v) - q_{i+1}(v)\bigr),$$

with indices taken cyclically. The interval evaluation supplies lower endpoints $\underline{\Delta}_{B,i}$ such that

$$\underline{\Delta}_{B,i} \le \Delta_{B,i}(v) \qquad (v \in B).$$

The cyclic-order certificate requires

$$\underline{\Delta}_{B,i} > 0$$

for the determinant rows used by the prescribed order. The order is part of the certificate data; the determinant lower bounds certify that this order remains positively oriented on $B$. The shoelace estimate is applied only after this order certificate is accepted.

### 6.3 Shoelace lower endpoints

For the accepted order, set

$$S_B(v) = \sum_{i=1}^{m_B} \bigl(x_i(v)y_{i+1}(v) - x_{i+1}(v)y_i(v)\bigr), \qquad q_{m_B+1} = q_1$$

by cyclic convention. The accepted orientation gives

$$\operatorname{area}(W_B(v)) = \frac{1}{2} S_B(v).$$

The interval computation supplies a lower endpoint $\underline{S}_B$ satisfying

$$\underline{S}_B \le S_B(v) \qquad (v \in B).$$

Thus

$$\operatorname{area}(W_B(v)) \ge \frac{1}{2}\underline{S}_B.$$

If $\frac{1}{2}\underline{S}_B \ge \tau$, Lemma 1 gives $A(v) \ge \tau$ on $B$.

## 7 The certified lower-bound theorem

### 7.1 Finite-cover proposition

**Proposition 1** (Finite-cover implication)**.** *Let $\mathcal{F}$ be a finite family of parameter domains such that*

$$\Omega_{\mathrm{adm}} \subseteq \bigcup_{B \in \mathcal{F}} B.$$

*Assume that for every $B \in \mathcal{F}$ there is a certified number $L_B$ satisfying*

$$A(v) \ge L_B \ge \tau \qquad (v \in B).$$

*Then $A(v) \ge \tau$ for every $v \in \Omega_{\mathrm{adm}}$.*

*Proof.* Let $v \in \Omega_{\mathrm{adm}}$. By the finite-cover relation, $v \in B$ for at least one $B \in \mathcal{F}$. The local certificate on that domain gives

$$A(v) \ge L_B \ge \tau.$$

Since $v$ was arbitrary, the inequality holds on $\Omega_{\mathrm{adm}}$. ∎

### 7.2 Certified lower-bound theorem

**Theorem 1** (Certified lower-bound theorem)**.** *Assume that the finite certificate satisfies the cover condition, the local lower-bound conditions, the witness-containment conditions on* $\mathcal{F}_{\mathrm{wit}}$*, and the interval orientation and shoelace lower-bound conditions stated above. Then*

$$A(v) \geq \tau \qquad (v \in \Omega_{\mathrm{adm}}).$$

*Proof.* The certificate supplies a finite cover $\mathcal{F}$ of $\Omega_{\mathrm{adm}}$. For $B \in \mathcal{F}_{\mathrm{sup}}$, the supporting records supply the local inequality $A(v) \geq L_B \geq \tau$ on $B$. For $B \in \mathcal{F}_{\mathrm{wit}}$, Lemma 1 and the interval orientation and shoelace conditions give $A(v) \geq L_B^{\mathrm{wit}} \geq \tau$ on $B$. Thus every domain in $\mathcal{F}$ satisfies the local hypothesis of Proposition 1, and the theorem follows. $\square$

### 7.3 Convex universal-cover consequence

**Corollary 1** (Convex universal-cover consequence)**.** *Under the hypotheses of Theorem 1,*

$$\alpha_{\mathrm{cvx}} \geq \tau = 0.83201.$$

*Proof.* Let $K \in \mathcal{U}_{\mathrm{cvx}}$. By the normalization principle of Section 2.3, choose $v \in \Omega_{\mathrm{adm}}$ such that $H(v) \subseteq K$. Then

$$\mathrm{area}(K) \geq \mathrm{area}(H(v)) = A(v).$$

Theorem 1 gives $A(v) \geq \tau$. Hence $\mathrm{area}(K) \geq \tau$. Taking the infimum over $K \in \mathcal{U}_{\mathrm{cvx}}$ proves the claim. $\square$

## 8 Conclusion

The certificate proves $A(v) \geq \tau$ on the admissible normalized three-test-set domain, with $\tau = 0.83201$. The proof is finite-cover based: the admissible domain is covered by finitely many parameter domains, each domain carries a local lower-bound certificate, and the witness domains are certified by inner-witness polygons whose areas are bounded below by interval orientation and shoelace estimates. The resulting convex universal-cover consequence is $\alpha_{\mathrm{cvx}} \geq 0.83201$.

The statement proved here is the convex lower-bound consequence obtained within the stated Brass–Sharifi three-test-set certificate model. It is not a statement about the unrestricted nonconvex problem, nor is it a proof-assistant formalization.

**Artifact availability.** The certificate records and verification code accompanying this paper are available from the project repository.[1]


## Acknowledgements

The author used ChatGPT as an auxiliary editorial tool. All mathematical claims, computations, certificate judgments, and final text are the responsibility of the author.


[1] https://github.com/Sheldon-Anderson/lebesgue-universal-cover-new-lower-bound-certificate

# References


[1] P. Brass and M. Sharifi. A lower bound for Lebesgue's universal cover problem. *International Journal of Computational Geometry & Applications*, 15(5):537–544, 2005.

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[3] P. Gibbs. An upper bound for Lebesgue's covering problem. *arXiv:1810.10089*, 2018.

[4] R. E. Moore. *Interval Analysis*. Prentice-Hall, Englewood Cliffs, NJ, 1966.

# A Finite certificate data

Table A.1: Finite certificate components.

| Component | Quantity | Role |
|---|---|---|
| parameter domains | 356,816 | cover of $\Omega_{\mathrm{adm}}$ |
| directed interval records | 41,261 | supporting local bounds |
| tensor records | 8,751 | auxiliary local bounds |
| bridge records | 282 | residual local bounds |
| witness domains | 16 | witness-domain local bounds |

Table A.2: Finite certificate conditions.

| Group | Assertion | Method |
|---|---|---|
| OB-A | $\Omega_{\mathrm{adm}} \subseteq \bigcup_{B \in \mathcal{F}} B$ | finite-cover check |
| OB-B | supporting domains satisfy local bounds | interval records |
| OB-C | witness domains satisfy local bounds | containment and shoelace intervals |
| OB-D | every used local record has evidence | individual evidence records |
| OB-E | interval endpoints clear $\tau$ | outward-rounded arithmetic |
| OB-F | the finite-cover implication is applied to the stated convex claim | final certificate check |

# B Witness-domain interval summaries

Table B.1: Witness-domain counts.

| Quantity | Value |
|---|---|
| witness domains | 16 |
| accepted terminal subdomains | 140 |
| failed terminal subdomains | 0 |
| witness point incidences | 2,112 |
| point-containment certificates | 2,112 / 2,112 |

Table B.2: Interval lower-bound summaries on the witness domains.

| Quantity | Value |
|---|---|
| minimum witness-domain area bound | 0.8642876791 |
| minimum witness-domain margin over $\tau$ | 0.0322776791 |
| minimum orientation determinant lower endpoint | $3.6637 \times 10^{-5}$ |

The minima in Table B.2 are taken only over the witness domains. They are not asserted to be the global infimum of $A(v)$ over $\Omega_{\mathrm{adm}}$; the theorem uses only that every accepted local record clears the threshold $\tau$.